mathematical machinery. Science wasn't about an objectively real world that existed independently of our perceptions, because anything that existed beyond perception—even a putatively "real" world—was simply metaphysics. Any statements that scientists made about nonobservable yet "real" things on the basis of their scientific theories were dismissed as unnecessary hypotheses, extraneous metaphysical baggage irrelevant to the true task of science. Electrons, for example, weren't real—they could not be seen. Only visible tracks in particle detectors like cloud chambers could be considered real, since that was all that could be directly perceived. Physicists certainly talked about electrons as if they were real, but this was merely a shorthand for their perceptions and not to be taken literally. Science was an instrument for predicting perceptions, nothing more. This view of science came to be known as *instrumentalism.*

The positivists also held that scientists and philosophers should strive for "unity of science"—a single, consistent worldview, based on science and observation, with the different sciences forming a continuous and consistent whole. Biology should be grounded in chemistry, which should be grounded in physics, and so on. This seems relatively innocent and uncontroversial now, but, at the time, there were strong movements within the sciences that pushed against this. Physics and chemistry were at odds for much of the nineteenth century—chemists mostly believed in atoms, whereas physicists were often skeptical that atoms existed—and only in the first decades of the twentieth century did the two fields start to build a consistent picture of chemical interactions. And biology was still not totally on board. Some biologists of the time believed in *vitalism*—the idea that living organisms were not subject to the same laws of physics as inanimate matter, that there was something nonphysical in cell division and inheritance that defied thermodynamics. The positivists rejected this claim, and others like it, as meaninglessly vague metaphysics. Even philosophy itself was to be subsumed by the unity of science, according to the Vienna Circle manifesto: "*There is no such thing as philosophy as a basic or universal science alongside or above the various fields of the one empirical science.*" Philosophy, like the natural sciences, should be grounded in statements about observation and sensations.

Despite its emphasis on empiricism and logic, the Vienna Circle didn't limit its concerns to science and philosophy—the unity of science extended